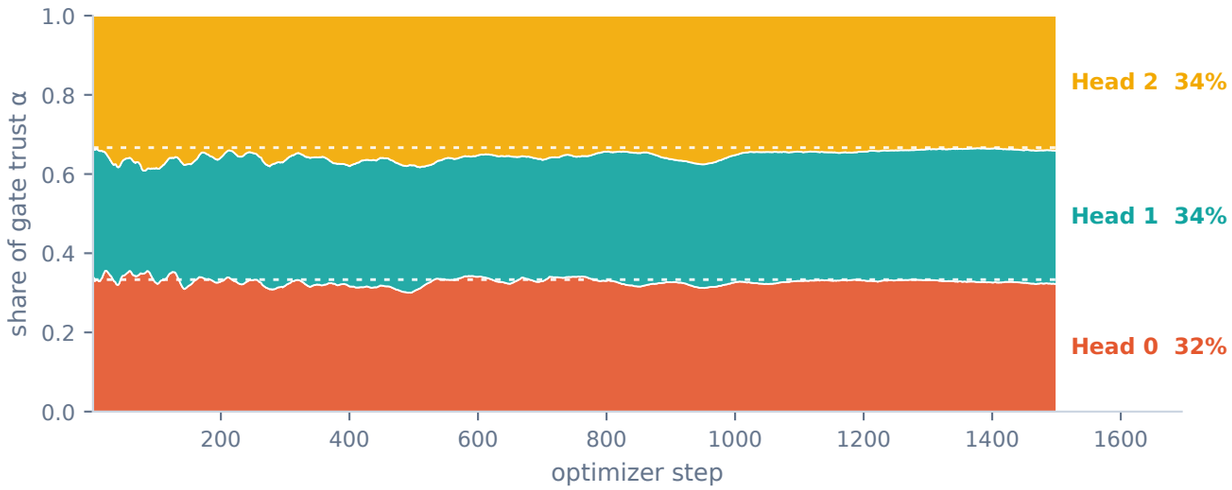
How the heads emerge and how the gate behaves

Heads start identical and are never told what to specialise in · first measurable divergence at step 22

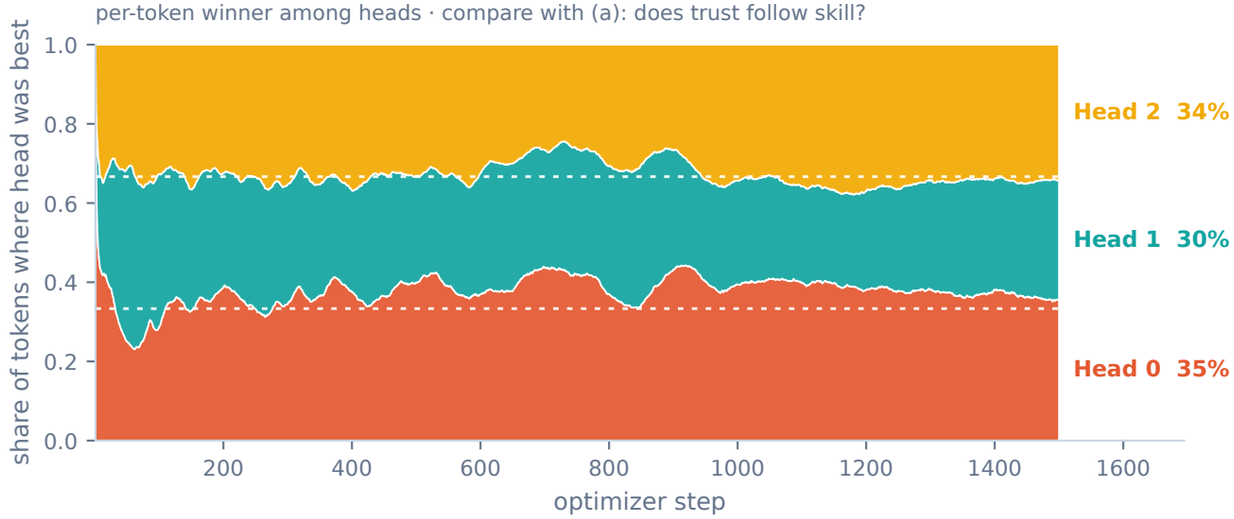
a Who the gate trusts

white dashes = perfectly uniform split · bands drifting from them = real arbitration



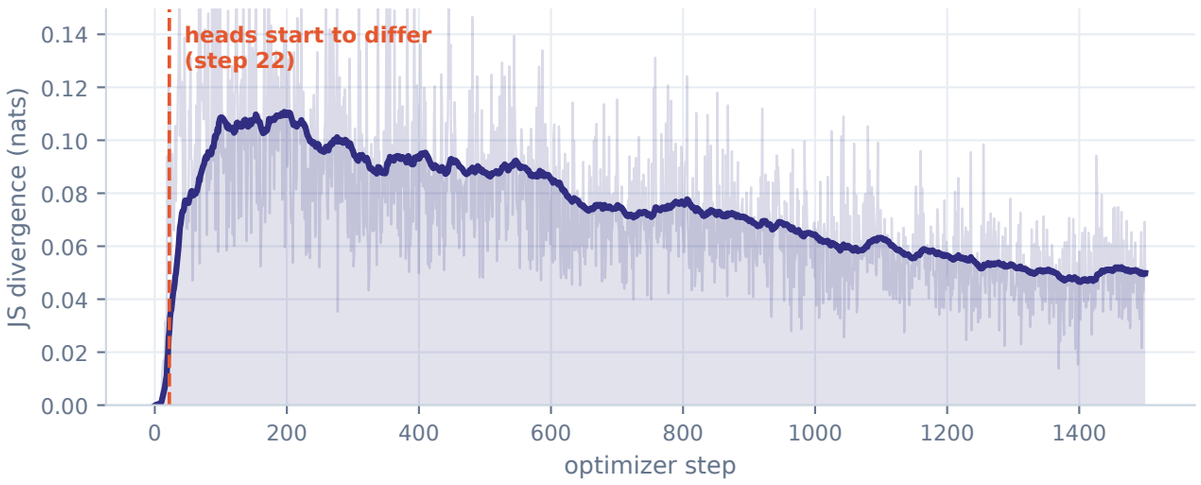
b Who is actually best

per-token winner among heads · compare with (a): does trust follow skill?



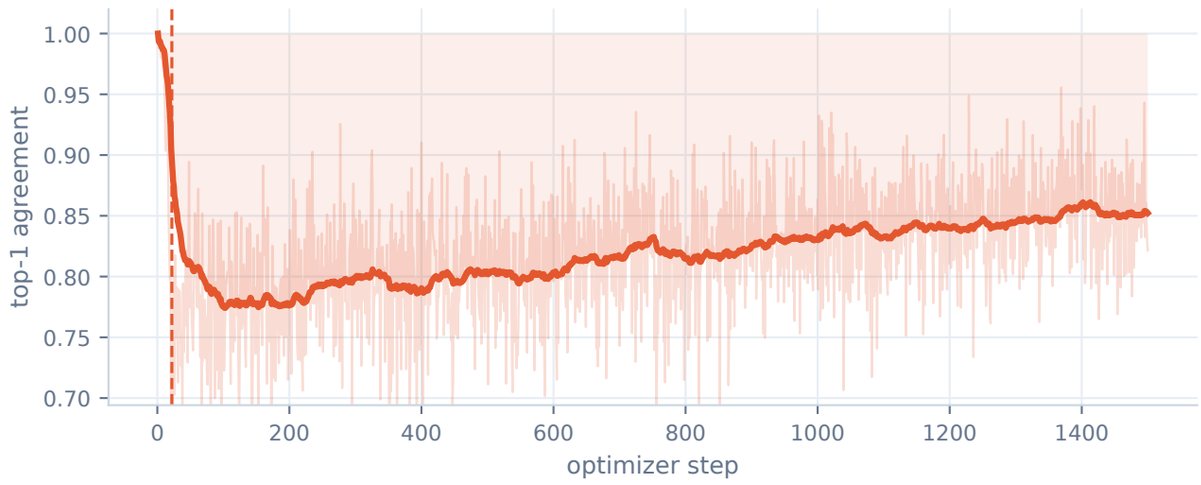
c Head disagreement (Jensen-Shannon)

0 = all heads output the same distribution · identical at init because LoRA B = 0



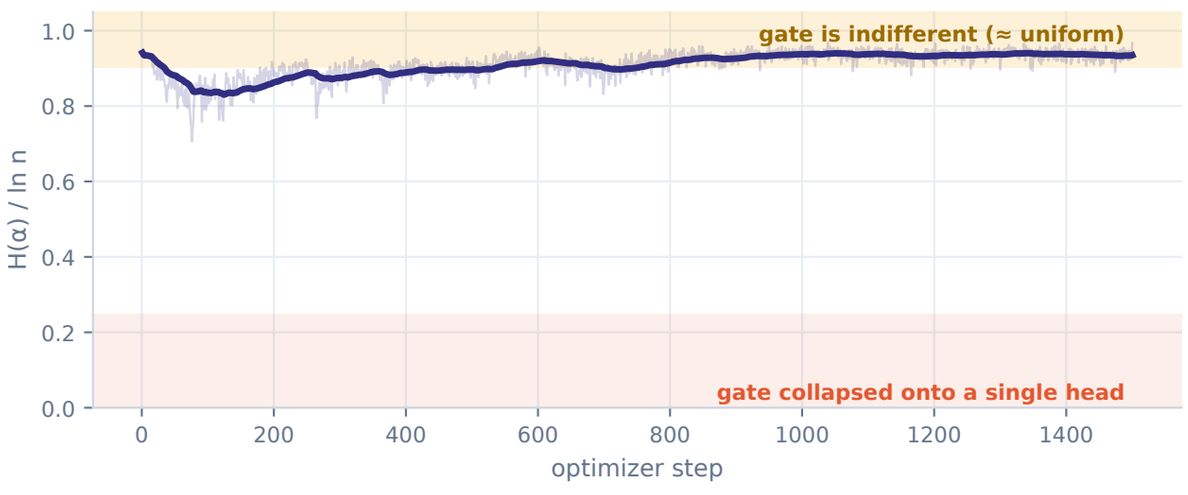
d Do the heads pick the same word?

share of tokens where every head's top-1 choice is identical · shaded = disagreement



e Is the gate actually deciding?

normalised entropy of α per token · the healthy zone is in between



f How far each head moved from the base model

‖B‖ of the LoRA update, all layers · a head that stays small is a dead head

